Drag of the Aconcagua rocket: CFD results and comparison with flight data

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Abstract

The drag coefficient of the Aconcagua sounding rocket was computed with OpenFOAM v2412 on a series of octree meshes, with a pressure-based solver below Mach 1 and a density-based central-upwind solver above it. The flight of the vehicle at IREC was reduced to a drag curve with an atmosphere fitted to the GPS and barometric data of the same flight. On the coarsest mesh, the subsonic drag at zero incidence agrees with the flight within 2 to 9 %, and decreases towards the flight value on the first refined mesh. The flight shows a transonic drag rise of a factor of two that a semi-empirical model does not predict. The supersonic computations are not yet converged: the central-upwind run started from the pressure-based solution shows a growth of the wall shear stress without limit. This report gives the results; the method is in the user guide ACG-CFD-UG-001.

1 Introduction

The aim of the work is the curve $C_D(M, \alpha)$ of the Aconcagua at the conditions of its flight, with an estimate of the discretization error, for trajectory simulation and for the comparison of two CFD solvers of different nature (OpenFOAM and Nektar++). The workflow, its parameters and the reasons for each choice are in the user guide (`flightModel/docs/guide/`). This report gives the results obtained up to October 19, 2026 and the open problems.

2 Computational setup

2.1 Geometry and meshes

The rocket is 2.955 m long with a body diameter $d = 0.151\,m$, a Von Kármán nose of 0.8 m, a boattail to 0.110 m, and four biconvex fins of constant thickness 11.7 mm with 1 mm edge radii and 3 mm root fillets. The meshes are half meshes (symmetry plane $z = 0$, fins at $45°$) made by cutting the full octree mesh. Only `maxCellSize` changes between meshes, by $\sqrt{2}$; the layer parameters keep the first cell thickness constant.

The octree does not refine uniformly: the number of cells grows by 1.95 from h_1 to h_2 and by 2.34 from h_2 to h_3, so the effective refinement ratios are 1.25 and 1.33. The measured first cell is 0.41 times the value of the layer rule, the same on all meshes, so the rule keeps t_1 constant ($y \approx 0.05\,mm$ at the cell centre).

Table 1: Meshes.

Mesh	`maxCellSize` [m]	Cells	Neg. vol.	Skewness	Non-orth.	t_1 [mm]
h_0	2.2627	1 074 798	0	2.37	65.7°	0.097
h_1	1.6	2 198 939	0	1.72	66.0°	0.100
h_2	1.1314	4 285 770	0	1.58	65.6°	0.096
h_3	0.8	10 020 768	0	1.41	65.9°	0.101

2.2 Solvers and flight conditions

$M < 1$: `rhoPimpleFoam` with the transonic pressure equation; $M > 1$: `rhoCentralFoam` (Kurganov flux, van Leer reconstruction). Both use local time stepping, k–ω SST with Spalding wall functions, and air as a perfect gas with Sutherland viscosity ($Pr = 0.690$ from the Eucken correction). The runs stop when the 1000-iteration moving mean of C_D is within 0.007 of the current value.

The first sweep (Section 4) used ISA conditions at the barometric altitude of the flight. The flight analysis (Section 3) showed an atmosphere about 20 K warmer than ISA; the sweep in progress uses the fitted atmosphere (Table 2).

Table 2: Flight conditions: first sweep (ISA) and current sweep (fitted atmosphere).

	ISA (first sweep)		Fitted atmosphere		
M	p_∞ [Pa]	T_∞ [K]	t [s]	p_∞ [Pa]	T_∞ [K]
1.8	80 127	275.56	4.1	78 354	295.82
1.2	62 346	262.72	8.5	60 547	280.15
0.8	49 800	251.72	14.6	47 509	266.17
0.6	42 766	244.54	19.8	40 911	257.90
0.3	36 113	236.79	29.0	34 613	248.96

2.3 Hardware

Two computers were used: a dual-socket workstation (2 × Xeon E5-2660 v4, 14 cores per socket, ECC memory, Open MPI 4.1) and a laptop (Ryzen 7 250, 8 cores, 28 GB without ECC, Open MPI 5.0). The workstation stopped twice under load because of a defective memory module and is out of service; the runs since 2026-10-15 are on the laptop. The memory need is about 3 GB for each million cells, so h_3 does not fit in the laptop.

3 Flight data

3.1 Atmosphere and speed

A fit of the hydrostatic profile with a constant lapse rate to 215 pairs of GPS height and barometric pressure on the descent gives $T_0 = 305.0\,K$ at the pad (896 m MSL), a lapse rate of 7.2 K/km and $p_0 = 90\,565\,Pa$, with an rms error of 0.3 % in $\ln p$. The atmosphere was approximately ISA + 20 K from the ground to 8 km; individual layers deviate from the fit by up to 7 K.

The speed of the flight computer, integrated, reaches 7837 m at apogee against 8083 m from the GPS: it converts pressure to height with ISA and is about 10 % low in the coast. The speed was therefore integrated from the axial accelerometer, anchored at $t = 18\,s$ to the barometric height with the fitted atmosphere. The integrated apogee is 8137 m (0.7 % above the GPS).

3.2 Drag curve

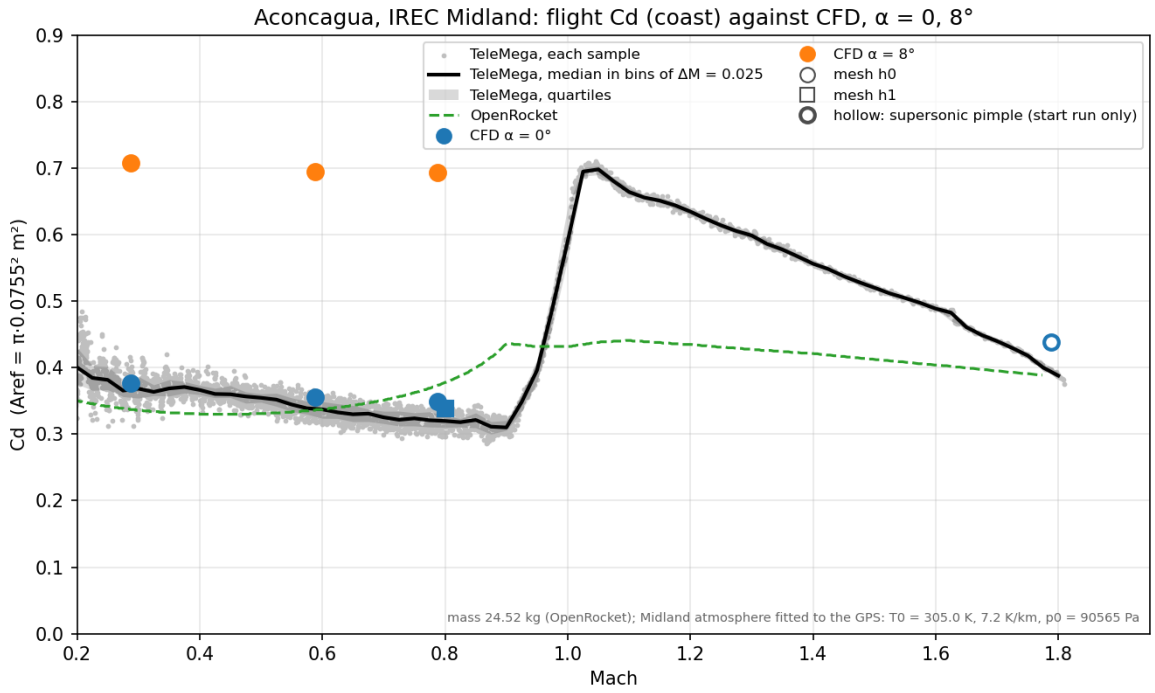


Figure 1: Flight C_D in the coast against the first sweep (ISA conditions) and OpenRocket.

With a post-burnout mass of 24.52 kg (OpenRocket), the flight C_D is nearly constant at 0.32–0.37 below $M = 0.9$, rises to about 0.70 at $M \approx 1.03$, and decreases to 0.39 at $M = 1.8$ (Figure 1). OpenRocket follows the subsonic values but gives only about 0.44 at the transonic peak.

3.3 Angle of attack

The vehicle rolled at about 4 revolutions per second in the coast. The mean lateral acceleration follows $\omega_{\text{roll}}^2 r$ with $r \approx 2.8\,cm$ (sensor off the axis). The component at the roll frequency gives $C_N = 0.01$–0.03, that is $\alpha < 0.2°$ throughout the coast with $C_{N\alpha} \approx 18\,/rad$. The angle of $16°$ was therefore removed from the sweep and replaced by $4°$.

4 Subsonic results

The first sweep ran nine conditions on h_0 and one on h_1 (ISA conditions). Coefficients are means over the last 1000 iterations, in wind axes; C_N and C_A are in body axes.

- At zero incidence, C_D is 2 to 9 % above the flight (Table 4), and the h_1 run is 3 % below the h_0 run, towards the flight value.
- The normal-force slope grows with incidence (vortex lift of the body and fins): $C_{N\alpha} \approx$ 18–19 /rad at 8°, 20–21 /rad at 16°. The centre of pressure is at 15.2–15.8 calibres from the nose, ahead of the fins ($x = 2.53\,m$), and moves forward with incidence.

Table 3: Coefficients of the first sweep.

Run	C_D	C_L	C_m	C_N	C_A	$C_{N\alpha}$ [1/rad]	x_{cp} [m]
M0.3, 0°, h_0	0.376	0.020	-0.38	–	0.376	–	–
M0.6, 0°, h_0	0.356	0.007	-0.14	–	0.356	–	–
M0.8, 0°, h_0	0.349	0.002	-0.03	–	0.349	–	–
M0.8, 0°, h_1	0.339	0.003	-0.06	–	0.339	–	–
M0.3, 8°, h_0	0.708	2.449	-39.6	2.524	0.361	18.1	2.37
M0.6, 8°, h_0	0.695	2.525	-41.1	2.597	0.337	18.6	2.39
M0.8, 8°, h_0	0.694	2.534	-41.3	2.606	0.334	18.7	2.39
M0.3, 16°, h_0	1.760	5.172	-83.2	5.456	0.266	19.5	2.30
M0.6, 16°, h_0	1.812	5.432	-88.2	5.721	0.245	20.5	2.33
M0.8, 16°, h_0	1.881	5.641	-92.1	5.941	0.253	21.3	2.34

- The axial force decreases with incidence (C_A from 0.36 to 0.27 at $M = 0.3$); the drag in wind axes rises because of the normal force.
- C_L at zero incidence is 0.02 at $M = 0.3$, small but not zero.
- Mean y^+ on the body walls: 2.4 ($M = 0.3$), 5.0 (0.6), 6.9 (0.8); on the fins 12 to 33.

Table 4: Zero-incidence drag: flight, CFD and OpenRocket.

M	Flight	CFD	CFD/flight	OpenRocket
0.3	0.369	0.376 (h_0)	1.02	0.336
0.6	0.337	0.356 (h_0)	1.05	0.337
0.8	0.320	0.349 (h_0), 0.339 (h_1)	1.09, 1.06	0.378
1.8	0.388	0.438 (pimple start run)	1.13	0.389

5 Supersonic computations

5.1 Start

From a uniform free stream, `rhoCentralFoam` failed at iterations 74 to 256 in three tests on h_1 at $M = 1.8$: the base acts as a piston and empties the region behind it, and the temperature there falls to zero. `rhoPimpleFoam`, with a temperature limit, survives the transient: the base pressure falls to 7900 Pa at iteration 305 and recovers to $p_{\text{base}}/p_\infty \approx 0.57$ by iteration 755. A central run started from this solution was stable over its ramp. The supersonic chains therefore start with a pimple run on the coarsest mesh.

5.2 Divergence of the central run

The central run started from the converged pimple solution increased C_D without limit, mostly through the wall shear stress, which tripled (Table 5). The run was stopped by a hardware failure at iteration 584, not by the solver. The flight value at this Mach number is 0.39. The cause is open; the candidates are the transfer of the turbulence fields, the wall treatment of the central template and the limiter at the wall. Two numerical defects found during this work are corrected: `mapFields` can fail with a segmentation fault when source and target are the

Table 5: Pimple start run against the central run started from it, h_0, $M = 1.8$.

	Pimple, converged	Central, iteration 550
C_D	0.437	0.894, increasing
Viscous force, x [N]	266	828
Pressure force, x [N]	445	604
Mean y^+ on the body	16.6	33.7

same mesh, which earlier went undetected; and under Open MPI 5 the process binding was silently ignored.

6 Computational cost

Without suspended intervals, the h_0 runs on the laptop cost 20–22 core-µs per cell and iteration (2.7 s per iteration with 8 processes), against 1.77 s per iteration with 14 processes on one socket of the workstation. The laptop is 1.5 times slower per iteration but cheaper per core. The instrumented runs record CPU cycles, effective clock, IPC and package energy: on h_0 with 8 bound processes the effective clock is 4.4 GHz and the IPC about 1, consistent with a memory-bound solver. These measurements are the basis of the cost comparison with Nektar++.

7 Second sweep: fitted atmosphere

The second sweep uses the fitted atmosphere and $\alpha = 0, 4, 8°$, with the measurement of the computer on each run. Table 6 gives the runs that finished.

Table 6: C_D of the second sweep (mean of the last 1000 iterations).

M	0°, h_0	4°, h_0	8°, h_0	0°, h_1	Flight
0.3	0.383	0.462	0.713	0.352	0.369
0.6	0.361	0.451	0.700	0.331	0.337
0.8	0.354	0.433	0.700	0.342	0.320

With the real atmosphere, C_D on h_0 is 1–2 % above the ISA sweep. On h_1 it is 3–8 % below h_0 and within -5 % to $+7$ % of the flight. $m08$-$a08$-h_1 gives 0.676.

Failure of the h_1 runs at incidence. Five of the six h_1 runs at $\alpha \neq 0$ stopped at iterations 2 to 7 with a floating point exception in the turbulence model. A repetition with the computer cold failed at the same iteration, so the failure is deterministic; a repetition without the mapped `rDeltaT` failed too. The cause: 7083 cells of h_1 near the fin tips are not inside h_0, so `mapFields` leaves them with the values of `0.orig`, and `0.orig` has no density field. These cells got ρ equal to the numerical value of p_∞, about 4×10^4, and the first pressure solution reached 3×10^9 Pa. With `0/rho` deleted after the mapping, the run continues with a minimum local time step of 4×10^{-8} s instead of 2×10^{-12} s.

Thermal state of the computer. In the long runs the CPU reached its 100 °C limit and one memory module 88.5 °C, above its 85 °C rating. The effective clock varied from 3.5 to 4.4 GHz

between runs with the same CPU profile, so the wall time of these runs is not a comparable cost without the cycle counts.

8 Reference values for verification

A replication of the first run of the guide (mesh h_0, $M = 0.8$, $\alpha = 0$, `pInf` 47 509 Pa, `Tinf` 266.17 K, `rhoPimpleFoam`) must give:

Table 7: Reference values of the first run.

Quantity	Value
Cells of h_0	1 074 798
Negative cell volumes	0
t_1 on the body walls	0.097 mm
C_D, mean of the last 1000 iterations	0.354
Mean y^+ on the body walls	6.2

A difference in C_D larger than the stop tolerance (0.007) shows a different geometry, mesh or setup. A small difference in the number of cells can come from another version of cfMesh.

9 Status and next steps

1. The second sweep (Section 7): repeat the five h_1 runs at incidence with the corrected `Allrun`, then h_2; h_3 waits for the workstation.
2. Mesh convergence and grid convergence index for C_D at each condition.
3. Diagnosis and correction of the central-run divergence, then the supersonic sweep and the transonic drag rise.
4. The same cases with Nektar++, with h and p refinement, and the comparison of accuracy against cost.

References

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[2] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, and P. E. Raad. Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130:078001, 2008.